



VP AIR TEST NEET (2025-26)

PART TEST - 03

DROPPER

DURATION: 180 Minutes

Date: 28-09-2025

M. MARKS: 720

ANSWER KEY

1. (3)	46. (2)	91. (2)	136. (2)
2. (2)	47. (1)	92. (1)	137. (1)
3. (1)	48. (4)	93. (3)	138. (2)
4. (2)	49. (3)	94. (1)	139. (4)
5. (3)	50. (2)	95. (2)	140. (2)
6. (2)	51. (2)	96. (3)	141. (1)
7. (1)	52. (1)	97. (3)	142. (1)
8. (1)	53. (4)	98. (2)	143. (4)
9. (2)	54. (4)	99. (4)	144. (3)
10. (2)	55. (4)	100. (1)	145. (1)
11. (1)	56. (3)	101. (1)	146. (2)
12. (2)	57. (2)	102. (1)	147. (3)
13. (2)	58. (2)	103. (4)	148. (2)
14. (1)	59. (4)	104. (4)	149. (3)
15. (3)	60. (2)	105. (1)	150. (4)
16. (3)	61. (3)	106. (2)	151. (3)
17. (2)	62. (3)	107. (3)	152. (3)
18. (1)	63. (1)	108. (1)	153. (2)
19. (1)	64. (4)	109. (3)	154. (2)
20. (2)	65. (4)	110. (3)	155. (1)
21. (1)	66. (1)	111. (3)	156. (1)
22. (3)	67. (4)	112. (2)	157. (3)
23. (1)	68. (1)	113. (4)	158. (4)
24. (3)	69. (3)	114. (1)	159. (2)
25. (3)	70. (3)	115. (1)	160. (4)
26. (3)	71. (1)	116. (1)	161. (2)
27. (4)	72. (3)	117. (1)	162. (1)
28. (1)	73. (3)	118. (1)	163. (1)
29. (2)	74. (3)	119. (2)	164. (2)
30. (4)	75. (3)	120. (1)	165. (1)
31. (3)	76. (3)	121. (1)	166. (2)
32. (2)	77. (4)	122. (2)	167. (4)
33. (4)	78. (1)	123. (2)	168. (3)
34. (3)	79. (1)	124. (1)	169. (2)
35. (3)	80. (4)	125. (3)	170. (2)
36. (4)	81. (4)	126. (4)	171. (3)
37. (3)	82. (1)	127. (2)	172. (1)
38. (3)	83. (1)	128. (1)	173. (1)
39. (1)	84. (3)	129. (3)	174. (3)
40. (1)	85. (2)	130. (3)	175. (1)
41. (2)	86. (2)	131. (4)	176. (4)
42. (4)	87. (1)	132. (1)	177. (2)
43. (3)	88. (1)	133. (1)	178. (3)
44. (2)	89. (2)	134. (3)	179. (1)
45. (4)	90. (1)	135. (1)	180. (3)

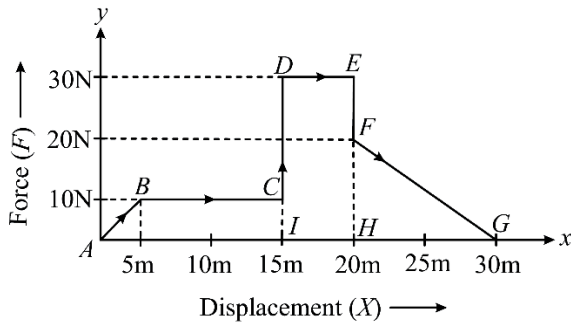
HINTS & SOLUTIONS

1. (3)

From law of inertia.

(NCERT Ed. 2025-26, 11th, Page No. 52)

2. (2)



Work done = Area under the force displacement graph

= Area of trapezium ABCI + Area of rectangle IDEH + Area of triangle FHG

$$= \left[\frac{1}{2} (10 + 15) 10 \right] + [5 \times 30] + \left[\frac{1}{2} \times 10 \times 20 \right]$$

$$= 125 + 150 + 100 = 375 \text{ J.}$$

(NCERT Ed. 2025-26, 11th, Page No. 76)

3. (1)

$$S = h + 2e^2h + 2e^4h + \dots$$

$$S = h + 2h[e^2 + e^4 + e^6 + \dots]$$

$$S = h + 2h \left[\frac{e^2}{1 - e^2} \right]$$

$$S = \frac{h(1 + e^2)}{(1 - e^2)}$$

(NCERT Ed. 2025-26, 11th, Page No. 84)

4. (2)

$$F_1 t_1 = F_2 t_2$$

(NCERT Ed. 2025-26, 11th, Page No. 73)

5. (3)

$$W = (3\hat{i} + c\hat{j} + 2\hat{k}) \cdot (-4\hat{i} + 2\hat{j} + 3\hat{k}) = 6J$$

$$-12 + 2c + 6 = 6 \Rightarrow c = 6$$

(NCERT Ed. 2025-26, 11th, Page No. 74)

6. (2)

Work done = $\int \vec{F} \cdot d\vec{s}$ or area under force displacement curve.

$$\text{For } A = W = \frac{1}{2} \times 4 \times 100 - 2 \times 50$$

$$\Rightarrow 200 - 100 = 100 \text{ J}$$

$$\text{For } B \Rightarrow 5 + 15 - 15 + 10 = 15 \text{ J}$$

$$\text{For } C \Rightarrow 5 + 10 + \frac{15}{2} - 15 = 7.5 \text{ J}$$

$$\text{For } D \Rightarrow 9 + 4.5 = 13.5 \text{ J}$$

(NCERT Ed. 2025-26, 11th, Page No. 76)

7. (1)

$$f_A = \mu m_A g$$

$$= 0.1 \times 10 \times 10 = 10 \text{ N}$$

$$f_B = \mu m_B g$$

$$= 0.1 \times 20 \times 10 = 20 \text{ N}$$

Force on A

$$F_A = m_A a_A = 10 \times 12 = 120 \text{ N}$$

so total absorbed force

$$F_A + f_A + f_B$$

$$= 120 + 10 + 20 = 150 \text{ N}$$

$$\text{Now rest of force} = 200 - 150 = 50 \text{ N}$$

Accelerates to the body 20 kg

$$\therefore F_B = m_B a_B$$

$$50 = 20 a_B$$

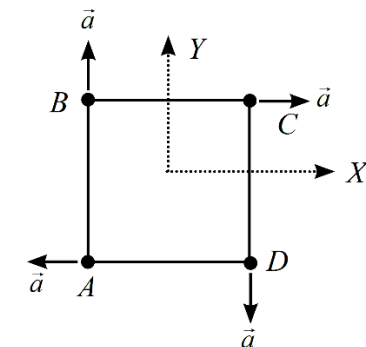
$$a_B = 2.5 \text{ m/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 65)

8. (1)

$$\vec{a}_{\text{cm}} = \frac{m_A \vec{a}_A + m_B \vec{a}_B + m_C \vec{a}_C + m_D \vec{a}_D}{m_A + m_B + m_C + m_D}$$

$$\vec{a}_{\text{cm}} = \frac{-ma\hat{i} + 2ma\hat{j} + 3ma\hat{i} - 4ma\hat{j}}{10m}$$



$$\vec{a}_{\text{cm}} = \frac{a}{5} \hat{i} - \frac{a}{5} \hat{j} = \frac{a}{5} (\hat{i} - \hat{j})$$

(NCERT Ed. 2025-26, 11th, Page No. 99)

9. (2)

According to the conservation of energy, kinetic energy at A + potential energy at B

$$\Rightarrow 0 + mgh = \frac{1}{2} mv^2 + 0$$

$$\text{or } v^2 = 2gh = 2 \times 9.8 \times 0.20$$

$$(\because h = \text{radius} = 20 \text{ cm} = 0.2 \text{ m})$$

According to work - energy theorem, Work done on the ball = change in kinetic energy

$$= \frac{1}{2}mv^2 - (0)^2 = \frac{1}{2} \times \frac{2}{1000} \times 2 \times 9.8 \times 0.2$$

$$= 3.92 \times 10^{-3} \text{ J} = 3.92 \text{ mJ}$$

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10. (2)

$$v = 72 \text{ km/hour} = 20 \text{ m/sec}$$

$$\theta = \tan^{-1} \left(\frac{v^2}{rg} \right) = \tan^{-1} \left(\frac{20 \times 20}{20 \times 10} \right)$$

$$= \tan^{-1}(2)$$

(NCERT Ed. 2025-26, 11th, Page No. 63)

11. (1)

$$W_G + W_f = 0 - 0$$

$$10 \times 1 + W_f = 0$$

$$10 - \mu mgx = 0$$

$$10 = (0.2)(10)x,$$

$$x = 5 \text{ m}$$

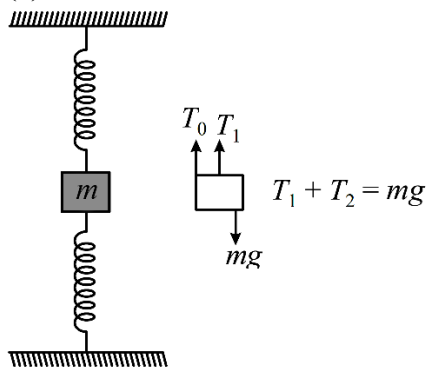
(NCERT Ed. 2025-26, 11th, Page No. 82)

12. (2)

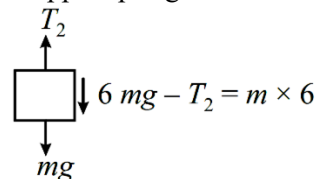
It is case of head on collision between objects of same masses with one of them (C and D system) being at rest initially the velocity exchange.

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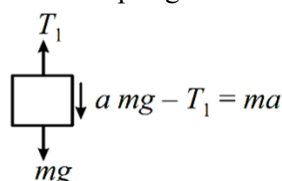
13. (2)



If upper spring is cut:



If lower spring is cut:



Adding (i) and (ii)

$$2mg - \{T_1 + T_2\} = m(a + 6)$$

$$2mg - mg = m(a + 6)$$

$$mg = m(a + 6)$$

$$g = a + 6a = 4 \text{ m/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 58)

14. (1)

$$F = T_1$$

$$T_2 = 2T_1$$

$$T_3 = 2T_2 = 4T_1$$

$$T_4 = T_1 + T_2 + T_3$$

$$= T_1 + 2T_1 + 4T_1 = 7T_1$$

$$7T_1 = 100$$

$$T_1 = \frac{100}{7} \text{ N}$$

$$T_2 = \frac{200}{7} \text{ N}$$

$$T_3 = \frac{400}{7} \text{ N}$$

$$T_4 = 100 \text{ N}$$

(NCERT Ed. 2025-26, 11th, Page No. 58)

15. (3)

The velocity of wall would not change after collision

$$e = \frac{-(-2 - V_1)}{-2 - 5} = \frac{1}{2}$$

$$\Rightarrow -2 - V_1 = \frac{7}{2}$$

$$\Rightarrow V_1 = -\frac{11}{2} \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 84)

16. (3)

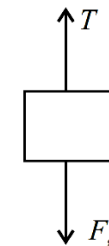
$$\text{Net force} = \vec{F}_1 + \vec{F}_2 = 8\hat{i} + 4\hat{j} + 4\hat{k}$$

$$\vec{a} = \frac{\vec{F}}{m} = 2\hat{i} + \hat{j} + \hat{k}$$

(NCERT Ed. 2025-26, 11th, Page No. 54)

17. (2)

Statement-I



When the elevator is moving with uniform speed,

$$T = F_g$$

Statement-II

When the elevator is going down with increasing speed, its acceleration is in the downward direction.

$$W - N = \frac{W}{g} \times a$$

$$N = W \left(1 - \frac{a}{g} \right) \text{ i.e. less than weight.}$$

Force exerted by the floor is less than the weight W .

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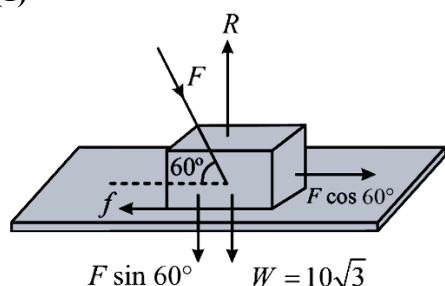
18. (1)

For exerted by ball on wall
= rate of change in momentum of ball

$$= \frac{mv - (-mv)}{t} = \frac{2mv}{t}.$$

(NCERT Ed. 2025-26, 11th, Page No. 56)

19. (1)



$$f = \mu R$$

$$F \cos 60^\circ = \mu(W + F \sin 60^\circ)$$

Substituting

$$\mu = \frac{1}{2\sqrt{3}} \quad \& \quad W = 10\sqrt{3} \quad \text{we get } F = 20\text{N}$$

(NCERT Ed. 2025-26, 11th, Page No. 60)

20. (2)

Loss in gravitational potential energy
= Elastic potential energy stored in compressed spring

$$\Rightarrow mg(h+x) = \frac{1}{2} kx^2$$

(NCERT Ed. 2025-26, 11th, Page No. 80)

21. (1)

$$v_0 = at_0$$

$$\Rightarrow a = \frac{v_0}{t_0}$$

$$\therefore v = \frac{v_0}{t_0} \cdot t$$

$$\Rightarrow w = \Delta k = k_f - k_i$$

$$\Rightarrow \frac{1}{2} M \frac{v_0^2}{t_0^2} t^2$$

(NCERT Ed. 2025-26, 11th, Page No. 73)

22. (3)

Work energy theorem.

$$\Sigma W = \Delta KE$$

$$\Rightarrow mgh + W_f = \frac{1}{2} mv^2 - 0$$

Work done by friction force

$$\text{i.e., } W_f = \frac{1}{2} mv^2 - mgh$$

$$= \frac{1}{2} \times 1 \times 4 - 1 \times 10 \times 1$$

$$W_f = -8\text{J}$$

(NCERT Ed. 2025-26, 11th, Page No. 73)

23. (1)

Downward retardation means upward acceleration.

$$g' = g + a$$

$$\text{Now } t = \sqrt{\frac{2L}{g \sin \theta}} \quad \text{or} \quad t = \sqrt{\frac{2L}{(g+a) \sin \theta}}$$

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24. (3)

At the highest point, $V \geq \sqrt{gR}$.

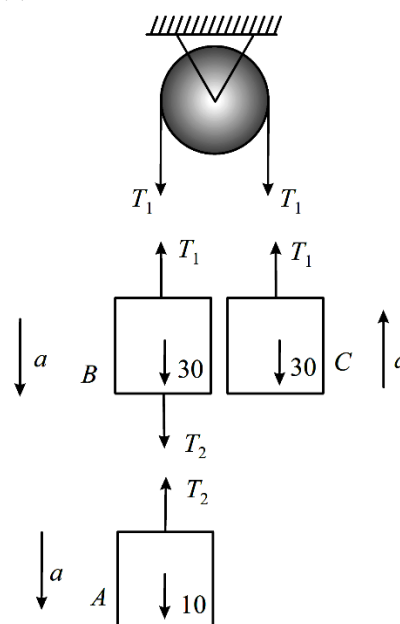
(NCERT Ed. 2025-26, 11th, Page No. 79)

25. (3)

$$P = \vec{F} \cdot \vec{v} = 50 - 30 + 120 = 140 \text{ J/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 83)

26. (3)



$$10 - T_2 = 1a$$

[Newton's II law for A]

$$T_2 + 30 - T_1 = 3a$$

[Newton's II law for B]

$$T_1 - 30 = 3a$$

[Newton's II law for C]

$$\Rightarrow a = \frac{g}{7} \Rightarrow T_2 = \frac{6g}{7}$$

(NCERT Ed. 2025-26, 11th, Page No. 65)

27. (4)

$$(W.D)_{\text{by friction}} + (W.D)_{\text{by spring}} = \Delta K =$$

$$K_f - K_i = 0 - K_i$$

$$-0.25 \times 1 \times 10 \times 4 - \frac{1}{2} \times 2.75 \times 4^2$$

$$= -\frac{1}{2} \times 1 \times v^2$$

$$v = 8 \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 82)

28. (1)

We know:

$$P = Fv$$

at any instant 't'

$$F = m \frac{dv}{dt}$$

$$\Rightarrow P = mv \frac{dv}{dt} = mv \left(\frac{dv}{ds} \right) \times \left(\frac{ds}{dt} \right) = mv^2 \frac{dv}{ds}$$

$$\text{or } \int_0^s ds = \int_v^{2v} \frac{mv^2 dv}{P} \Rightarrow s = \frac{7mv^3}{3P}$$

(NCERT Ed. 2025-26, 11th, Page No. 83)

29. (2)

Using work energy theorem, work done by all forces = change in kinetic energy,

$$\Rightarrow W_g + W_{Fr} + W_s = \Delta KE$$

$$5 \times 10 \times 5 - 0.5 \times 5 \times 10 \times x - \frac{1}{2} Kx^2 = 0 - 0$$

$$2x^2 + x - 10 = 0$$

$$x = 2 \text{ m}$$

(NCERT Ed. 2025-26, 11th, Page No. 82)

30. (4)

Due to the same mass of A and B as well as due to elastic collision velocities of spheres get interchanged after the collision.

(NCERT Ed. 2025-26, 11th, Page No. 84)

31. (3)

At equilibrium,

$$\frac{dU}{dr} = 0 \Rightarrow \frac{d}{dr} \left[\frac{A}{r^{10}} \right] - \frac{d}{dr} \left[\frac{B}{r^5} \right] = 0$$

$$\Rightarrow -10Ar^{-10-1} - B(-5)r^{-5-1} = 0$$

$$\Rightarrow -10Ar^{-11} + 5Br^{-6} = 0$$

$$10Ar^{-11} = 5Br^{-6}$$

$$\Rightarrow r^5 = \frac{2A}{B}$$

$$\therefore r = \left(\frac{2A}{B} \right)^{\frac{1}{5}}$$

(NCERT Ed. 2025-26, 11th, Page No. 77)

32. (2)

Applying principle of conservation of linear momentum, we get

$$mu = (M + m)V$$

$$10^{-2} \times 2 \times 10^2 = 1 \times V$$

$$V = 2 \text{ m/s}$$

$$h = \frac{V^2}{2g} = 0.2 \text{ m}$$

(NCERT Ed. 2025-26, 11th, Page No. 84)

33. (4)

$$\text{Centre of mass are } r_{\text{cm}} = \frac{h}{4} = \frac{40}{4}$$

$$= 10 \text{ cm}$$

(NCERT Ed. 2025-26, 11th, Page No. 97)

34. (3)

$$V_{\text{cm}} = \frac{M_1 V_1 + M_2 V_2}{M_1 + M_2}$$

$$= \frac{200 \times 10\hat{i} + 500(3\hat{i} + 5\hat{j})}{700} = 5\hat{i} + \frac{25}{7}\hat{j}$$

(NCERT Ed. 2025-26, 11th, Page No. 99)

35. (3)

$$mg - \frac{3}{4}mg = ma$$

[Newton's II law for man]

$$\Rightarrow a = \frac{g}{4}$$

(NCERT Ed. 2025-26, 11th, Page No. 61)

36. (4)

$$\text{Since, } P = \frac{mgh}{t}$$

$$\frac{P_1}{P_2} = \frac{\frac{m_1 gh}{t_1}}{\frac{m_2 gh}{t_2}} = \frac{m_1 t_2}{t_1 m_2}$$

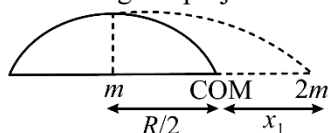
$$\frac{3\sqrt{x}}{\sqrt{x}+1} = \frac{300 \times 2}{5 \times 50} = \frac{12}{5}$$

$$12\sqrt{x} + 12 = 15\sqrt{x} \Rightarrow x = 16$$

(NCERT Ed. 2025-26, 11th, Page No. 83)

37. (3)

Centre of mass hits the ground at the position where original projectile would have landed.



$$m \times \frac{R}{2} = 2mx_1 \Rightarrow x_1 = \frac{R}{4}$$

$$\therefore \text{Distance} = R + \frac{R}{4} = \frac{5R}{4}$$

(NCERT Ed. 2025-26, 11th, Page No. 100)

38. (3)

$$mu = (m + m)V$$

$$V = \frac{u}{2} \quad \dots(1)$$

Initial KE = Final P.E

$$\frac{1}{2} \times (2m)V^2 = (2m)gh$$

$$\frac{1}{2}V^2 = gh$$

From (1),

$$\frac{1}{2} \times \left(\frac{u}{2}\right)^2 = gh$$

$$\frac{1}{2} \times \frac{u^2}{4} = gh$$

$$h = \frac{u^2}{8g}$$

(NCERT Ed. 2025-26, 11th, Page No. 84)

39. (1)

$$x_{\text{COM}} = \frac{(\sigma\pi a^2)a - \sigma\pi\left(\frac{a}{2}\right)^2 \cdot \left(\frac{3a}{2}\right)}{\sigma\pi a^2 - \sigma\pi\left(\frac{a}{2}\right)^2}$$

$$= \frac{a - \frac{3a}{8}}{1 - \frac{1}{4}} = \frac{5a}{6}$$

(NCERT Ed. 2025-26, 11th, Page No. 98)

40. (1)

$$f = \mu N$$

$$49 = \mu \times 10 \times 9.8$$

$$\mu = \frac{1}{2} = 0.5$$

$$\theta = \tan^{-1}(0.5)$$

(NCERT Ed. 2025-26, 11th, Page No. 61)

41. (2)

$$\int_0^v dv = \int_1^2 \frac{1}{2}(t^2 - 1)dt$$

$$v = \frac{1}{2} \left[\frac{t^3}{3} - t \right]_1^2$$

$$v = \frac{2}{3} \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 54)

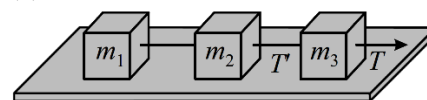
42. (4)

Centre of mass may or may not lie at the geometric centre, depending upon the mass distribution of the object.

Centre of gravity and centre of mass do not always coincide.

(NCERT Ed. 2025-26, 11th, Page No. 112)

43. (3)



$$F = ma$$

$$T' = (m_1 + m_2) \times \frac{T}{m_1 + m_2 + m_3}$$

(NCERT Ed. 2025-26, 11th, Page No. 65)

44. (2)

$$a_{\text{max}} = \mu_s g = 0.3 \times 9.8 = 2.94 \text{ m/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 61)

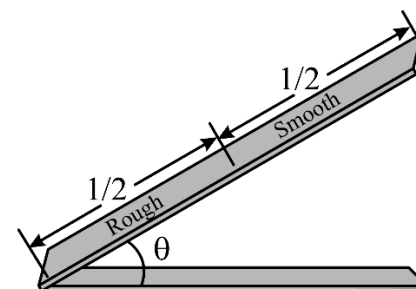
45. (4)

For upper half

$$v^2 = u^2 + 2a \frac{l}{2} = 2(g \sin \theta) \frac{l}{2} = gl \sin \theta$$

For lower half

$$\Rightarrow 0 = u^2 + 2g(\sin \theta - \mu \cos \theta) \frac{l}{2}$$



$$\Rightarrow -gl \sin \theta = gl(\sin \theta - \mu \cos \theta)$$

$$\Rightarrow \mu \cos \theta = 2 \sin \theta \Rightarrow \mu = 2 \tan \theta$$

(NCERT Ed. 2025-26, 11th, Page No. 82)

46. (2)

Let, 1 kg solvent (H_2O) $\therefore$ mole of $\text{KOH} = 1.5$ mass of $\text{KOH} = 1.5 \times 56 = 84 \text{ g}$ mass of solution = $1000 + 84 = 1084 \text{ g}$

$$\text{Molarity} = \frac{1.5}{\left(\frac{1084}{0.8}\right) \times 10^{-3}} = 1.11 \text{ M}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 04)

47. (1)

Lattice enthalpy of NaCl means the energy required to separate one mole of solid NaCl into one mole of $\text{Na}^+(\text{g})$ and one mole of $\text{Cl}^-(\text{g})$

(NCERT Ed. 2025-26 11th Part-I Page No. 107)

48. (4)

- Vapour pressure a liquid increases non-linearly with increasing temperature.
- Relative lowering of vapour pressure is a colligative property but vapour pressure is not a colligative property.

(NCERT Ed. 2025-26 12th Part-I Page No. 12)

49. (3)

- $\text{O}=\dot{\text{N}}\rightarrow\text{O}$: odd electron molecule
- $\text{H}-\text{Be}-\text{H}$: incomplete octet molecule
- $\text{Cl}-\ddot{\text{P}}(\text{Cl})_2$: complete octet
- F_6S : 12 electron around S-atom (expanded octet)

(NCERT Ed. 2025-26 11th Part-I Page No. 105)

50. (2)

$$\text{Mass \% of X} = \frac{4}{4+36} \times 100 = 10\%$$

(NCERT Ed. 2025-26 12th Part-I Page No. 02)

51. (2)

Intermolecular H-bonding is present in HF .(NCERT Ed. 2025-26 11th Part-I Page No. 132)

52. (1)

$$p_{\text{CO}_2} = K_{\text{H}} X_{\text{CO}_2} \approx K_{\text{H}} \times \frac{n_{\text{CO}_2}}{N_{\text{H}_2\text{O}}}$$

$$0.8 = 1.67 \times 10^3 \times \left(\frac{n_{\text{CO}_2}}{1000}\right)$$

$$\therefore n_{\text{CO}_2} = \frac{0.8 \times 1000}{18 \times 1.67 \times 10^3} = 0.026$$

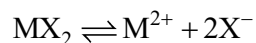
$$m_{\text{CO}_2} = 0.026 \times 44 = 1.14 \text{ g}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 07)

53. (4)

molecule NH_3 HF H_2O $\mu(\text{D})$ 1.47 1.78 1.85(NCERT Ed. 2025-26 11th Part-I Page No. 112)

54. (4)



$$i = 1 + (3-1)\alpha = 1 + 2\alpha$$

$$i = \frac{M_{\text{normal}}}{M_{\text{observed}}} \Rightarrow 1 + 2\alpha = \frac{170}{70}$$

$$\alpha = 0.714 \quad \therefore \% \alpha = 71.4$$

(NCERT Ed. 2025-26 12th Part-I Page No. 24)

55. (4)

More the charge and lesser the size of cation, more will be its polarising power.

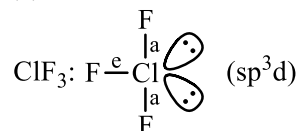
(NCERT Ed. 2025-26 11th Part-I Page No. 112)

56. (3)

A pressure cooker reduces cooking time for food because boiling point of water involved in cooking is increased.

(NCERT Ed. 2025-26 12th Part-I Page No. 16)

57. (2)



axial bond length is more than equatorial bond length.

(NCERT Ed. 2025-26 11th Part-I Page No. 117)

58. (2)

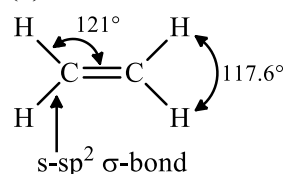
$$\frac{p^0 - p_s}{p^0} = X_{\text{solute}} \approx \frac{n}{N}$$

$$0.2 = \frac{12}{m} \times \frac{18}{90}$$

$$m = 12$$

(NCERT Ed. 2025-26 12th Part-I Page No. 15)

59. (4)

(NCERT Ed. 2025-26 11th Part-I Page No. 123)

60. (2)
- $\text{CHCl}_3 + \text{CH}_3\text{COCH}_3$: Maximum boiling azeotrope (negative deviation)
 - $\text{C}_2\text{H}_5\text{OH} + \text{water}$: Minimum boiling azeotrope (positive deviation)
 - K_b : Molal elevation constant or ebullioscopic constant
 - K_f : Molal depression constant or cryoscopic constant

(NCERT Ed. 2025-26 12th Part-I Page No. 14, 17, 19)

61. (3)
- Positive overlap is the in phase overlap orbital. It results due to overlapping of orbitals having same sign and orientation in space.

(NCERT Ed. 2025-26 11th Part-I Page No. 119)

62. (3)
- Commercial bleaching solution contains 3.62 mass percentage of sodium hypochlorite in water.

(NCERT Ed. 2025-26 12th Part-I Page No. 02)

63. (1)
- $$\text{B}_2(10e^-): (\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\pi 2p_x)^1 (\pi 2p_y)^1$$
- B_2 contains two unpaired electron so paramagnetic in nature.

(NCERT Ed. 2025-26 11th Part-I Page No. 131)

64. (4)
- Amalgam of mercury with sodium is a type of solid solution.

(NCERT Ed. 2025-26 12th Part-I Page No. 02)

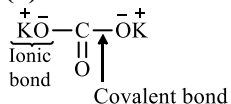
65. (4)
- Due to small size and high electronegativity only HF can form KHF_2 type salt.

(NCERT Ed. 2025-26 11th Part-I Page No. 131)

66. (1)
- Bromoethane and chloroethane form ideal solution on mixing.

$$\therefore \Delta_{\text{mix}}H=0, \Delta_{\text{mix}}V=0, \Delta_{\text{mix}}G<0 \text{ and } \Delta_{\text{mix}}S>0$$

(NCERT Ed. 2025-26 12th Part-I Page No. 13)

67. (4)
- 

(NCERT Ed. 2025-26 11th Part-I Page No. 101, 102)

68. (1)
- Water boils at 373.15 K (100°C) because at this temperature the vapour pressure of water is 1.013 bar (1 atmosphere).

(NCERT Ed. 2025-26 12th Part-I Page No. 16)

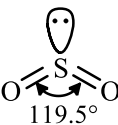
69. (3)
- Bond order of CO is 3.

(NCERT Ed. 2025-26 11th Part-I Page No. 109)

70. (3)
- More the value of $i \times m (\approx i \times M)$, more will be the boiling point of solution.

- Urea : $i \times M = 1 \times 0.01 = 0.01$
- Glucose : $i \times M = 1 \times 0.15 = 0.15$
- KNO_3 : $i \times M = 2 \times 0.10 = 0.2$
- Na_2SO_4 : $i \times M = 3 \times 0.02 = 0.06$

(NCERT Ed. 2025-26 12th Part-I Page No. 17)

71. (1)
- 

(NCERT Ed. 2025-26 11th Part-I Page No. 116)

72. (3)
- During osmosis only movement of solvent take place from low concentrated solution (NaCl) to high concentrated solution (AgNO_3), as a result molarity of NaCl solution increases and molarity of AgNO_3 solution decreases, also no ppt is obtain.

(NCERT Ed. 2025-26 12th Part-I Page No. 21)

73. (3)
- Due to absence of π -bond, resonance is not observed in NH_4^+ .

(NCERT Ed. 2025-26 11th Part-I Page No. 109)

74. (3)
- $$C_1 = C_2$$
- $$\frac{\left(\frac{5}{M}\right)}{100 \times 10^{-3}} = \frac{\left(\frac{2.5}{60}\right)}{100 \times 10^{-3}}$$
- $$\therefore M = \frac{5 \times 60}{2.5} = 120 \text{ g mol}^{-1}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 22)

75. (3)
- C_2 consists two pi bonds. It is due to presence of four electrons in two pi molecular orbitals.

(NCERT Ed. 2025-26 11th Part-I Page No. 130)

76. (3)
- $$p_{\text{Benzene}} = 80 \times \frac{1}{4} = 20 \text{ torr}$$
- $$p_{\text{toluene}} = 24 \times \frac{3}{4} = 18 \text{ torr}$$

$$Y_{\text{toluene}} = \frac{p_{\text{toluene}}}{p_{\text{total}}} = \frac{18}{20+18} = 0.47$$

(NCERT Ed. 2025-26 12th Part-I Page No. 11)

77. (4)
- Average bond length: $\text{C}-\text{C} > \text{C}-\text{N} > \text{N}-\text{O} > \text{O}-\text{H}$.

(NCERT Ed. 2025-26 11th Part-I Page No. 108)

78. (1)
- Higher the K_H at a given pressure, the lower is the solubility of the gas in the liquid.
 - K_H : Ar > CH₂=CH-Cl > CH₄ > HCHO
- (NCERT Ed. 2025-26 12th Part-I Page No. 08)

79. (1)
- Formal charges help in the selection of the lowest energy structure from a number of possible Lewis structures for a given species.
 - Ionic bonds will be formed more easily between elements with comparatively low ionization enthalpies and elements with comparatively high negative value of electron gain enthalpy.
- (NCERT Ed. 2025-26 11th Part-I Page No. 106, 105)

80. (4)
- 0.9% (mass/volume) NaCl solution is called normal saline solution.
 - Naphthalene (non-polar) does not dissolve in water (polar).
- (NCERT Ed. 2025-26 12th Part-I Page No. 06, 22)

81. (4)
- CO: $\text{C} \equiv \text{O}$
 - O₃: $\text{O}=\text{O} \rightarrow \text{O}$
 - NO₃⁻: $\text{O}=\text{N}-\text{O}^-$
 $\downarrow$
O
 - NO: $\text{N} \equiv \text{O}$
- (NCERT Ed. 2025-26 11th Part-I Page No. 105)

82. (1)
- 10⁶ g of solution contain 15 g CHCl₃.
- $$\therefore \% \text{ by mass} = \frac{15}{10^6} \times 100 = 1.5 \times 10^{-3}$$
- (NCERT Ed. 2025-26 12th Part-I Page No. 28)

83. (1)
-
- No. of σ -bonds = 9
- (NCERT Ed. 2025-26 11th Part-I Page No. 120)

84. (3)
- $$2\text{C}_6\text{H}_5\text{COOH} \xrightarrow{\text{C}_6\text{H}_6} (\text{C}_6\text{H}_5\text{COOH})_2$$
- (i) 1 0
- (f) $1 - \alpha$ $\frac{\alpha}{2}$
- $$i = 1 - \alpha + \frac{\alpha}{2} = 1 - \frac{\alpha}{2}$$
- $$i = 1 - \frac{0.6}{2} = 0.7$$

$$\Delta T_f = i \times K_f \times \text{molality}$$

$$1 = \frac{0.7 \times 5.12 \times \left(\frac{x}{122}\right)}{70 \times 10^{-3}}$$

$$x = 2.38 \text{ g}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 24)

85. (2)
- More the charge on cation or anion, more will be lattice energy.
- (NCERT Ed. 2025-26 11th Part-I Page No. 107)

86. (2)
- $$p_t = p_A^0 \chi_A + p_B^0 \chi_B$$
- $$400 = p_A^0 (0.8) + p_B^0 (0.2) \quad \dots (1)$$
- $$450 = p_A^0 (0.2) + p_B^0 (0.8) \quad \dots (2)$$
- Apply $4 \times \text{eq (2)} - 3 \times \text{eq (1)}$
- $$p_B^0 = 466.66 \text{ mm of Hg}$$
- (NCERT Ed. 2025-26 12th Part-I Page No. 10)

87. (1)
- I₃⁻: $\left[\text{I} \cdots \text{I} \cdots \text{I} \right]^-$ linear (sp^3)
 - BF₄⁻: $\left[\begin{array}{c} \text{F} \\ | \\ \text{B} \\ | \\ \text{F} \end{array} \right]^-$ tetrahedral (sp^3)
 - NH₂⁻: $\left[\begin{array}{c} \text{N} \\ | \\ \text{H} \end{array} \right]^-$ bent (sp^3)
 - H₃O⁺: $\left[\begin{array}{c} \text{O} \\ | \\ \text{H} \end{array} \right]^+$ trigonal pyramidal (sp^3)
- (NCERT Ed. 2025-26 11th Part-I Page No. 115)

88. (1)
- Molality is independent of temperature as it does not involve volume term.
 - Reverse osmosis is used in desalination of sea water.
- (NCERT Ed. 2025-26 12th Part-I Page No. 05, 23)

89. (2)
- $$\text{O}_2^-(17 e^-): (\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2p_z)^2 (\pi 2p_x^2 \equiv \pi 2p_y^2) (\pi^* 2p_x^2 \equiv \pi^* 2p_y^1)$$
- (NCERT Ed. 2025-26 11th Part-I Page No. 131)

90. (1)
- Initial mass of HNO₃ = $0.5 \times 0.8 \times 63 = 25.2 \text{ g}$
- Final mass of HNO₃ = $25.2 - 10.2 = 15 \text{ g}$
- $$\text{Final molarity} = \frac{\left(\frac{15}{63}\right)}{400 \times 10^{-3}} = 0.59 \text{ M}$$
- (NCERT Ed. 2025-26 12th Part-I Page No. 04)

91. (2)

In the Mendelian dihybrid cross, 9:3:3:1 ratio is obtained for the phenotype at F_2 . The yellow and green colour segregated in a 3:1 ratio. Round and wrinkled seed shape also segregated in a 3:1 ratio; just like in a monohybrid cross. These results were identical to those that he got when he made separate monohybrid crosses between yellow and green seeded plants and between round and wrinkled seeded plants. The progeny of the F_1 generation resembled one of the parents. Also, the genes of the two traits assort independently.

(NCERT Ed. 2025-26, 12th, Page No. 62)

92. (1)

Mendel published his work on inheritance of characters in 1865 but for several reasons, it remained unrecognised till 1900. Firstly, communication was not easy (as it is now) in those days and his work could not be widely publicised. Secondly, his concept of genes (or factors, in Mendel's words) as stable and discrete units that controlled the expression of traits and, of the pair of alleles which did not 'blend' with each other, was not accepted by his contemporaries. Thirdly, Mendel's approach of using mathematics to explain biological phenomena was totally new and unacceptable to many of the biologists of his time. Finally, though Mendel's work suggested that factors (genes) were discrete units, he could not provide any physical proof for the existence of factors or say what they were made of.

(NCERT Ed. 2025-26, 12th, Page No. 65)

93. (3)

Klinefelter's Syndrome: This genetic disorder is caused due to the presence of an additional copy of X-chromosome resulting into a karyotype of 47, XXY.

Turner's Syndrome: Such a disorder is caused due to the absence of one of the X chromosomes, i.e., 45 with XO.

Haemophilia: X-linked recessive disorder caused due to mutation in X chromosome.

Myotonic dystrophy: Autosome-linked dominant disorder caused due to mutation in an autosome.

(NCERT Ed. 2025-26, 12th, Page No. 76)

94. (1)

There are instances where a single gene can exhibit multiple phenotypic expression. Such a gene is called a pleiotropic gene. The underlying

mechanism of pleiotropy in most cases is the effect of a gene on metabolic pathways which contribute towards different phenotypes. An example of this is the disease phenylketonuria, which occurs in humans. The disease is caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase (single gene mutation). This manifests itself through phenotypic expression characterised by mental retardation and a reduction in hair and skin pigmentation.

(NCERT Ed. 2025-26, 12th, Page No. 69)

95. (2)

Since both male and female can get affected from the shown disorder, it cannot be a sex-linked disorder. Since there are carrier individuals in the family, the disorder must be of recessive type. Let's represent the trait as A, in which AA, A'A and A'A', respectively represent, normal, carrier and affected individuals. Considering parents from F_2 , the Punnett square will be:

	A'	A
A'	A'A'	A'A
A	A'A	AA

Only 25% of the progeny in F_3 will be absolutely normal, being neither a carrier nor affected.

(NCERT Ed. 2025-26, 12th, Page No. 73)

96. (3)

The law of Segregation is based on the fact that the alleles do not show any blending and that both the characters are recovered as such in the F_2 generation. In incomplete dominance for flower colour in snapdragon, the F_2 resulted in the following ratio 1 (RR) Red : 2 (Rr) Pink : 1 (rr) White. The recovery of red and white flowers proves that the alleles segregated.

(NCERT Ed. 2025-26, 12th, Page No. 60)

97. (3)

Occasionally, a single gene product may produce more than one effect.

For example, starch synthesis in pea seeds is controlled by one gene. It has two alleles (B and b). Starch is synthesised effectively by BB homozygotes and therefore, large starch grains are produced. In contrast, bb homozygotes have lesser efficiency in starch synthesis and produce smaller starch grains. After maturation of the seeds, BB seeds are round and the bb seeds are wrinkled. Heterozygotes produce round seeds, and so B seems to be the dominant allele. But, the starch grains

produced are of intermediate size in Bb seeds. So if starch grain size is considered as the phenotype, then from this angle, the alleles show incomplete dominance.

(NCERT Ed. 2025-26, 12th, Page No. 62)

98. (2)

The sequence of steps of an artificial hybridization experiment, as performed by Mendel in pea, is: Choosing two true-breeding lines [E], Removal of anthers (Emasculation) [C], followed by Transfer of pollen (Pollination) [A], which leads to seed formation that is collected [B], and then grown [D].

(NCERT Ed. 2025-26, 12th, Page No. 55)

99. (4)

The work of Mendel and others who followed him gave us an idea of inheritance patterns. However the nature of those 'factors' which determine the phenotype was not very clear. "As these 'factors' represent the genetic basis of inheritance, understanding the structure of genetic material and the structural basis of genotype and phenotype conversion became the focus of attention in biology for the next century."

(NCERT Ed. 2025-26, 12th, Page No. 51)

100. (1)

Genes which code for a pair of contrasting traits are known as alleles, i.e., they are slightly different forms of the same gene. The phenotype is the observable physical characteristic of an organism (e.g., tall or dwarf). Homozygous means an organism has two identical alleles for a trait (e.g., TT or tt). An organism with two different alleles (e.g., Tt) is called heterozygous. A dominant allele expresses itself in both the homozygous (e.g., TT) and heterozygous (e.g., Tt) states. An allele that only expresses itself in the homozygous state (e.g., tt) is called a recessive allele.

(NCERT Ed. 2025-26, 12th, Page No. 56)

101. (1)

Sutton and Boveri argued that the pairing and separation of a pair of chromosomes would lead to the segregation of a pair of factors they carried. Sutton united the knowledge of chromosomal segregation with Mendelian principles and called it the chromosomal theory of inheritance. Following this synthesis of ideas, experimental verification of the chromosomal theory of inheritance by Thomas Hunt Morgan and his colleagues, led to discovering the basis for the variation that sexual reproduction produced. Sturtevant used the frequency of recombination between gene pairs on the same

chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome. Henking (1891) could trace a specific nuclear structure all through spermatogenesis in a few insects, and it was also observed by him that 50 per cent of the sperm received this structure after spermatogenesis, whereas the other 50 per cent sperm did not receive it. Henking gave a name to this structure as the X body but he could not explain its significance.

(NCERT Ed. 2025-26, 12th, Page No. 67)

102. (1)

Klinefelter's Syndrome : This genetic disorder is also caused due to the presence of an additional copy of X-chromosome resulting into a karyotype of 47, XXY. Such an individual has overall masculine development, however, the feminine development (development of breast, i.e., Gynaecomastia) is also expressed. Such individuals are sterile.

Failure of segregation of chromatids during cell division cycle results in the gain or loss of a chromosome(s), called aneuploidy.

(NCERT Ed. 2025-26, 12th, Page No. 76)

103. (4)

To determine the genotype of a tall plant at F₂, Mendel crossed the tall plant from F₂ with a dwarf plant. This he called a test cross. In a typical test cross an organism showing a dominant phenotype (and whose genotype is to be determined) is crossed with the recessive parent.

(NCERT Ed. 2025-26, 12th, Page No. 58)

104. (4)

Since the children had different groups, it is possible only when one parent is heterozygous for blood group A and other for B.

	I^A	i
I^B	$I^A I^B$	$I^B i$
i	$I^A i$	ii

(NCERT Ed. 2025-26, 12th, Page No. 61)

105. (1)

If you look around you will find that there are many traits which are not so distinct in their occurrence and are spread across a gradient. For example, in humans we don't just have tall or short people as two distinct alternatives but a whole range of possible heights. Such traits are generally controlled by three or more genes and are thus called as polygenic traits. In a polygenic trait the phenotype reflects the contribution of each allele, i.e., the effect of each allele is additive.

(NCERT Ed. 2025-26, 12th, Page No. 69)

106. (2)

Pea seed colour shows Mendelian inheritance in which yellow colour is dominant over green. In a cross between a pure yellow-seeded pea plant and a pure green-seeded one, all F_1 offspring will have yellow seeds. Therefore, the percentage of green-colored seeds is 0%. The F_2 generation exhibits a classic Mendelian 3:1 phenotypic ratio, with three-fourths of the seeds being yellow. Therefore, the percentage of yellow-colored seeds is 75%.

The inheritance of flower colour in the dog flower (snapdragon or *Antirrhinum sp.*) is a good example to understand incomplete dominance. In a cross between true-breeding red-flowered (RR) and true-breeding white-flowered plants (rr), the F_1 (Rr) was pink (100%). When the F_1 was self-pollinated the F_2 resulted in the following ratio 1 (RR, 25%) Red: 2 (Rr, 50%) Pink: 1 (rr, 25%) White.

(NCERT Ed. 2025-26, 12th, Page No. 60)

107. (3)

Contrasting Traits Studied by Mendel in Pea

S.No.	Characters	Contrasting Traits
1.	Stem height	Tall/dwarf
2.	Flower colour	Violet/white
3.	Flower position	Axial/terminal
4.	Pod shape	Inflated/constricted
5.	Pod colour	Green/yellow
6.	Seed shape	Round/wrinkled
7.	Seed colour	Yellow/green

(NCERT Ed. 2025-26, 12th, Page No. 54)

108. (1)

In the case of co-dominance the F_1 generation resembles both parents. A good example is different types of red blood cells that determine ABO blood grouping in human beings. ABO blood groups are controlled by the gene I . The plasma membrane of the red blood cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the gene. The gene (I) has three alleles I^A , I^B and i . The alleles I^A and I^B produce a slightly different form of the sugar while allele i does not produce any sugar. Because humans are diploid organisms, each person possesses any two of the three I gene alleles. I^A and I^B are completely dominant over i , in other words when I^A and i are present only I^A expresses (because i does not produce any sugar), and when I^B and i are present I^B expresses. But when I^A and I^B are present together they both express their own types of sugars: this is because of co-dominance. Hence red blood cells have both A and B types of sugars. Since there are three different alleles, there are six

different combinations of these three alleles that are possible, and therefore, a total of six different genotypes of the human ABO blood types. Four phenotypes i.e. A, B, AB and O are possible.

(NCERT Ed. 2025-26, 12th, Page No. 61)

109. (3)

Grasshopper is an example of XO type of sex determination in which the males have only one X-chromosome besides the autosomes, whereas females have a pair of X-chromosomes.

(NCERT Ed. 2025-26, 12th, Page No. 70)

110. (3)

Thalassemia: This genetic disorder affects the body's ability to produce hemoglobin, the protein in red blood cells that carries oxygen. The lack of functional hemoglobin leads to anemia and other complications related to red blood cells. So, it is an RBC disorder.

Haemophilia : This sex linked recessive disease, which shows its transmission from unaffected carrier female to some of the male progeny has been widely studied. In this disease, a single protein that is a part of the cascade of proteins involved in the clotting of blood is affected.

(NCERT Ed. 2025-26, 12th, Page No. 74)

111. (3)

In a normal human, number of autosomes is 44 (22 pairs). Down's syndrome is a trisomy of 21st chromosome, thus having 45 autosomes. In Turner's syndrome and Klinefelter's syndrome individuals have 44 autosomes.

(NCERT Ed. 2025-26, 12th, Page No. 76)

112. (2)

The frequency of recombination between gene pairs on the same chromosome can be used as a measure of the distance between genes for mapping their position on the chromosome.

In the given question, the largest distance is between y and m (37.2 units). The sum of the other two distances ($y-w = 1.3$ and $w-m = 35.9$) is $1.3 + 35.9 = 37.2$. This confirms that gene 'w' is located between genes 'y' and 'm'.

(NCERT Ed. 2025-26, 12th, Page No. 67)

113. (4)

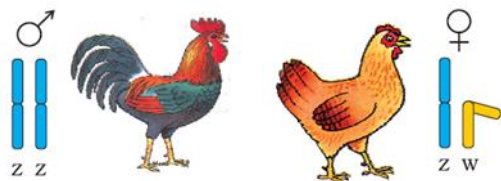
When Morgan hybridised yellow-bodied, white-eyed females to brown-bodied, red-eyed males and intercrossed their F_1 progeny, he observed that the two genes did not segregate independently of each

other and the F_2 ratio deviated very significantly from the 9:3:3:1 ratio (expected when the two genes are independent). Morgan attributed this due to the physical association or linkage of the two genes and coined the term linkage to describe this physical association of genes on a chromosome and the term recombination to describe the generation of non-parental gene combinations. Morgan and his group also found that even when genes were grouped on the same chromosome, some genes were very tightly linked (showed very low recombination) while others were loosely linked (showed higher recombination). For example he found that the genes white and yellow were very tightly linked and showed only 1.3 per cent recombination while white and miniature wing showed 37.2 per cent recombination.

(NCERT Ed. 2025-26, 12th, Page No. 67)

114. (1)

In birds, the total number of chromosome is same in both males and females. But two different types of gametes in terms of the sex-chromosomes, are produced by females, i.e., female heterogamety. In order to have a connection with the mechanism of sex determination described earlier, the two different sex chromosomes of a female bird has been designated to be the Z and W chromosomes. In these organisms the females have one Z and one W chromosome, whereas males have a pair of Z-chromosomes besides the autosomes.



In many birds, female has a pair of dissimilar chromosomes ZW and male two similar ZZ chromosomes

(NCERT Ed. 2025-26, 12th, Page No. 71)

115. (1)

Height in humans is a classic example of a polygenic trait. It is not controlled by a single gene but by the cumulative effect of multiple genes, along with environmental factors. Phenylketonuria is an inherited disorder where a single gene defect (in the gene for the enzyme phenylalanine hydroxylase) leads to multiple, seemingly unrelated phenotypic effects, such as mental retardation and reduced skin/hair pigmentation. When one gene influences

multiple traits, it is known as pleiotropy. Down's syndrome is caused by the presence of an extra copy of chromosome 21 (Trisomy 21). This change in the number of chromosomes makes it a chromosomal disorder. Myotonic dystrophy is an inherited condition that causes progressive muscle wasting and weakness. It is inherited as an autosomal dominant Mendelian disorder, meaning a mutation in just one copy of the responsible gene is enough to cause the disorder.

(NCERT Ed. 2025-26, 12th, Page No. 69)

116. (1)

A test cross is used to determine the unknown genotype of an organism that displays a dominant phenotype (in this case, round seeds). This is done by crossing the unknown organism with an organism that is homozygous recessive for the same trait (in this case, a pea plant with wrinkled seeds, genotype rr). Since the test plant has round seeds (the dominant trait), its genotype could be either Homozygous dominant (RR) or Heterozygous (Rr). Scenario 1: If the test plant is homozygous dominant (RR).

The cross would be: $RR \times rr$

The RR parent can only produce gametes with the R allele.

The rr parent can only produce gametes with the r allele.

All possible offspring would have the genotype Rr.

Phenotype of all offspring: 100% Round seeds.

Scenario 2: If the test plant is heterozygous (Rr).

The cross would be: $Rr \times rr$

The Rr parent produces two types of gametes: 50% with R and 50% with r.

The rr parent can only produce gametes with the r allele.

The offspring would have two possible genotypes: 50% Rr and 50% rr.

Phenotype of offspring: 50% Round seeds (Rr) and 50% wrinkled seeds (rr).

Since all the offsprings had round seed shape, it can be inferred that the test plant is homozygous dominant (RR).

(NCERT Ed. 2025-26, 12th, Page No. 59)

117. (1)

Law of Dominance:

- (i) Characters are controlled by discrete units called factors.
- (ii) Factors occur in pairs.

(iii) In a dissimilar pair of factors one member of the pair dominates (dominant) the other (recessive).

The law of dominance is used to explain the expression of only one of the parental characters in a monohybrid cross in the F_1 and the expression of both in the F_2 . It also explains the proportion of 3:1 obtained at the F_2 .

Law of Segregation:

This law is based on the fact that the alleles do not show any blending and that both the characters are recovered as such in the F_2 generation though one of these is not seen at the F_1 stage.

(NCERT Ed. 2025-26, 12th, Page No. 59)

118. (1)

The cross is between X^hX (carrier female) and XY (normal male).

	X^h	X
X	X^hX	XX
Y	X^hY	XY

Out of four progeny none is a haemophilic female.

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119. (2)

Mutation that arises due to change in a single base pair of DNA is known as point mutation. A classical example of such a mutation is sickle cell anemia. The defect is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule. The substitution of amino acid in the globin protein results due to the single base substitution at the sixth codon of the beta globin gene from GAG to GUG. The mutant haemoglobin molecule undergoes polymerisation under low oxygen tension causing the change in the shape of the RBC from biconcave disc to elongated sickle like structure.

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120. (1)

An analysis of traits in a several of generations of a family is called the pedigree analysis. In the pedigree analysis the inheritance of a particular trait is represented in the family tree over generations. This can be done for Mendelian disorders (e.g. cystic fibrosis) but not for chromosomal disorders (e.g. Turner's syndrome).

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121. (1)

In general, independent pairs of chromosomes segregate independently of each other. Also, one pair of gene segregates independently of another pair.

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122. (2)

In the F_2 generation of a Mendelian dihybrid cross, the phenotypic ratio of 9:3:3:1 was obtained, wherein: 9/16 will have both dominant traits (yellow, round) (parental) 3/16 will have the first dominant trait and the second recessive trait (yellow, wrinkled) (recombinant) 3/16 will have the first recessive trait and the second dominant trait (green, round) (recombinant) 1/16 will have both recessive traits (green, wrinkled) (parental)

Therefore a total of 10 parental and 6 recombinant progeny were obtained. Thus, the ratio of parental type to recombinant type progeny was: 10:6 or 5:3.

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123. (2)

A Punnett square can be used for a dihybrid cross, in the same way as used in monohybrid cross.

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124. (1)

During spermatogenesis among human males, two types of gametes are produced. 50 per cent of the total sperm produced carry the X-chromosome and the rest 50 per cent has Y-chromosome. Females, however, produce only one type of ovum with an X-chromosome. There is an equal probability of fertilisation of the ovum with the sperm carrying either X or Y chromosome. In case the ovum fertilises with a sperm carrying X-chromosome the zygote develops into a female (XX) and the fertilisation of ovum with Y-chromosome carrying sperm results into a male offspring. Thus, it is evident that it is the genetic makeup of the sperm that determines the sex of the child."

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125. (3)

Drosophila melanogaster complete their life cycle in about two weeks, and a single mating could produce a large number of progeny flies.

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126. (4)
The correct set of characters in a wild fruit fly is red eye, brown body and normal wing.

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127. (2)
Through artificial selection and domestication from ancestral wild cows, we have well-known Indian breeds, e.g., Sahiwal cows in Punjab. Mendel used pea plants for dihybrid crosses. The family pedigree of Queen Victoria shows a number of haemophilic descendents as she was a carrier of the disease.

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128. (1)
Phenylketonuria: This inborn error of metabolism is also inherited as the autosomal recessive trait. It is caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase (single gene mutation). The affected individual lacks an enzyme that converts the amino acid phenylalanine into tyrosine. As a result of this phenylalanine is accumulated and converted into phenylpyruvic acid and other derivatives. Accumulation of these in brain results in mental retardation. These are also excreted through urine because of its poor absorption by kidney.

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129. (3)
The sex determination in honey bee is based on the number of sets of chromosomes an individual receives. An offspring formed from the union of a sperm and an egg develops as a female (queen or worker), and an unfertilised egg develops as a male (drone) by means of parthenogenesis. This means that the males have half the number of chromosomes than that of a female. The females are diploid having 32 chromosomes and males are haploid, i.e., having 16 chromosomes. This is called as haplodiploid sex-determination system and has special characteristic features such as the males produce sperms by mitosis, they do not have father and thus cannot have sons, but have a grandfather and can have grandsons.

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130. (3)
 α Thalassemia is controlled by two closely linked genes HBA1 and HBA2 on chromosome 16 of each parent. While β Thalassemia is controlled by a single gene HBB on chromosome 11 of each parent.
Colour Blindness : This defect is due to mutation in certain genes present in the X chromosome.

Down's Syndrome : The cause of this genetic disorder is the presence of an additional copy of the chromosome number 21 (trisomy of 21).

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131. (4)
A fundamental principle of meiosis is that homologous chromosomes separate from each other during Meiosis I. This means that chromosome B and its homolog B' will be segregated into different daughter cells. The resulting gametes are haploid, meaning they contain only one chromosome from each homologous pair. Therefore, a normal gamete will have one chromosome from the A/A' pair and one from the B/B' pair. The orientation of the (A, A') pair is independent of the (B, B') pair. This leads to the following possible combinations in the gametes:
A and B
A and B'
A' and B
A' and B'

The combination B-B' is not usually present in a gamete. A gamete should not contain both members of a homologous pair. The presence of both B and B' in a single gamete would be the result of an error in meiosis.

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132. (1)
Inheritance is the process by which characters are passed on from parent to progeny. Some mutations involve changes in whole set of chromosomes (polyploidy) or change in a subset of chromosome number (aneuploidy).

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133. (1)
Colour blindness occurs in about 8 per cent of males and only about 0.4 per cent of females.

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134. (3)
Mendel observed that all the F_1 progeny plants were tall, like one of its parents; none were dwarf.

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135. (1)
The two sexes (male and female) were found to have a set of chromosomes which were common, and another set which was different. The chromosomes which were different in two sexes were named as sex chromosomes. The remaining set was named as autosomes.

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136. (2)

The Inspiratory Capacity (IC) is calculated using the following formula:

$$IC = VC - ERV$$

Where:

- VC is Vital Capacity (4800 ml)
- ERV is Expiratory Reserve Volume (1200 ml)

Plugging in the given values:

$$IC = 4800 \text{ ml} - 1200 \text{ ml} = 3600 \text{ ml}$$

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137. (1)

Blood coagulation can be initiated by two main pathways that converge on a common pathway:

- The extrinsic pathway is triggered by tissue trauma.
- The intrinsic pathway is activated by damage to the blood vessel itself. These two pathways show that the coagulation cascade is indeed complex and can be initiated by different types of injuries.
- While coagulation is a cascade of linked enzymatic reactions, the clotting factors are not always present in their active state. They are present in the plasma in an inactive form (e.g., prothrombin, factor X), and they must be sequentially activated in order for the cascade to proceed. If they were always active, blood would clot spontaneously.

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138. (2)

Oxygen (O_2) is utilised by the organisms to indirectly break down simple molecules like glucose, amino acids, fatty acids, etc., to derive energy to perform various activities.

Carbon dioxide (CO_2) is a toxic by-product of metabolism. It is not essential and must be eliminated from the body to prevent a dangerous drop in blood pH (respiratory acidosis).

Breathing is the physical process of inhaling air containing oxygen and exhaling air containing carbon dioxide. This gas exchange occurs between the lungs and the atmosphere. This is commonly known as respiration though technically different.

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139. (4)

Blood is a specialized fluid connective tissue, not a tissue fluid. Tissue fluid, also known as interstitial fluid, is the fluid that surrounds the cells in the tissues. Blood is contained within blood vessels and is distinct from the interstitial fluid that it exchanges

substances with. Therefore, classifying blood as a "tissue fluid" is fundamentally wrong.

While the protein concentration (6-8%) is stated correctly, the percentage of water in plasma is wrong. Plasma is composed of approximately 90-92% water. The figure 55% refers to the percentage of blood volume that is plasma, not the percentage of plasma volume that is water.

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140. (2)

Under normal physiological conditions, the partial pressure of oxygen at the tissue level is about 40 mm Hg, which matches Point B in the graph.

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141. (1)

List-I (Leucocyte)	List-II (Nucleus Shape)
Neutrophil	Multi-lobed
Eosinophil	Bilobed
Basophil	S-shaped
Monocyte	Kidney-shaped

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142. (1)

The central chemoreceptors are highly sensitive to changes in H^+ concentration in the cerebrospinal fluid (CSF), which is directly influenced by the PCO_2 in the arterial blood (CO_2 diffuses across the blood-brain barrier and forms carbonic acid, which dissociates into H^+ and HCO_3^-). However, they are not directly sensitive to changes in O_2 concentration. The response to changes in PO_2 is primarily mediated by the peripheral chemoreceptors (aortic and carotid bodies).

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143. (4)

The typical durations of the events in a single cardiac cycle (which lasts approximately 0.8 seconds):

- Joint Diastole: 0.4 seconds. During this phase, all four chambers of the heart are in a relaxed state. This is the longest phase of the cycle.
- Ventricular Systole: 0.3 seconds. This is the contraction phase of the ventricles, where blood is pumped out of the heart.
- Auricular Systole: 0.1 seconds. This is the contraction of the atria, which pushes the remaining blood into the ventricles. This is the shortest phase of the cycle.

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144. (3)

Capillaries are part of the closed circulatory system and form a continuous, closed network between arteries and veins. Open-ended vessels are characteristic of an open circulatory system, like that of insects.

The heart is a muscular organ that functions as the central pump in the circulatory system, propelling blood throughout the body.

The lymphatic system, with its fluid called lymph, plays a crucial role in the absorption of dietary fats and fat-soluble vitamins from the small intestine via specialized vessels called lacteals.

The Rh antigen is so named because it was first discovered in Rhesus monkeys, not Rhesus kangaroos.

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145. (1)

List-I	List-II
Passageways up to the terminal bronchioles	Filters, warms, and humidifies inspired atmospheric air
The respiratory zone	Actual site of O ₂ /CO ₂ diffusion across the respiratory membrane
Pulmonary Ventilation	The physical movement of air driven by pressure gradients
Mucus in respiratory tract	A defense mechanism that traps and removes inhaled pathogens and debris.

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146. (2)

The systemic circuit begins with the powerful contraction of the left ventricle, which has the most muscular wall of the heart chambers. This contraction pumps oxygenated blood into the aorta, the body's largest artery, for distribution to all tissues.

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147. (3)

- Lymph is a colorless to pale-yellow fluid. It is colorless precisely because it lacks Red Blood Cells (RBCs) and therefore also lacks the red pigment haemoglobin, which is contained within RBCs.
- It is formed from the fluid that bathes the body's tissues (interstitial fluid) after it drains into the lymphatic vessels.
- Lymph nodes, a key part of the lymphatic system, are rich in lymphocytes (B-cells and T-cells). These cells are critical for identifying and destroying pathogens, forming the basis of the body's adaptive immune response.
- Lymph is a vital carrier fluid. For example, fats absorbed in the small intestine are transported

via the lacteals (specialized lymphatic capillaries) in the form of chyle (a type of lymph). It also helps in the transport of hormones released by endocrine glands.

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148. (2)

The primary and most powerful stimulus that triggers the urge to breathe under normal physiological conditions is the increase in the partial pressure of carbon dioxide (pCO₂) in the blood. This rise in CO₂ leads to an increased concentration of hydrogen ions (H⁺) in the cerebrospinal fluid (CSF), which are detected by the central chemoreceptors in the medulla. While falling O₂ levels are also detected (by peripheral chemoreceptors), this stimulus is much weaker and only becomes significant during severe hypoxia (when pO₂ drops below ~60 mm Hg). Therefore, the rising CO₂ is the change that would first and most strongly lead to the urge to breathe.

When deoxygenated haemoglobin in the lungs binds with oxygen, it forms oxyhaemoglobin. Oxyhaemoglobin is a stronger acid than deoxygenated haemoglobin. This increased acidity causes it to release the H⁺ ions it was buffering. These released H⁺ ions then combine with bicarbonate ions (HCO₃⁻) to form carbonic acid (H₂CO₃), which is then converted to CO₂ and water. This process facilitates the release and exhalation of CO₂ from the blood in the lungs.

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149. (3)

Blood leaving the pulmonary capillaries and entering the pulmonary veins is fully oxygenated and has a pO₂ of about 104 mm Hg, which is in equilibrium with the alveolar air. However, the blood that is ultimately pumped by the left ventricle into the systemic circulation (systemic arterial blood) has a slightly lower pO₂ of about 95 mm Hg. The lungs themselves are living tissues and require their own oxygenated blood supply. This is provided by the bronchial arteries. After this blood supplies oxygen to the lung tissues (like the bronchi), it becomes deoxygenated. A significant portion of this deoxygenated bronchial venous blood drains directly into the pulmonary veins, mixing with the highly oxygenated blood coming from the alveoli.

This mixing of a small amount of deoxygenated blood with a large amount of oxygenated blood is called a physiological shunt. It effectively "dilutes" the concentration of oxygen in the blood that finally reaches the left atrium and is then passed to the left ventricle. This is the precise reason why the pO₂ drops from 104 mm Hg to the standard 95 mm Hg observed in systemic arteries.

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150. (4)

The first heart sound, 'lub', is produced by the closing of the atrioventricular (AV) valves—the mitral valve and the tricuspid valve. This sound marks the beginning of ventricular systole, as the ventricles start to contract and prevent backflow of blood into the atria.

The second heart sound, 'dub', is produced by the closing of the semilunar valves—the aortic and pulmonary valves. This sound marks the end of ventricular systole and the beginning of ventricular diastole (the relaxation phase) as blood pressure drops in the ventricles.

Hypertension (high blood pressure) is defined as a blood pressure reading consistently at or above 140/90 mmHg. It is a major risk factor for cardiovascular diseases and can cause severe damage to vital organs like the brain (leading to stroke), kidneys (leading to kidney failure), and heart (leading to heart attack).

An electrocardiogram (ECG) is a non-invasive procedure that measures the electrical activity of the heart, not its mechanical pumping action. It records the electrical impulses generated by the heart muscle to assess its rhythm and function.

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151. (3)

"Fully saturated" means haemoglobin is carrying its maximum load (4 oxygen molecules). It cannot bind to any more.

The movement of CO₂ from tissues into blood is by passive diffusion, not active transport (which requires energy).

Carbaminohaemoglobin is a combination of haemoglobin and carbon dioxide. When it breaks apart, it releases haemoglobin and CO₂, not oxygen. Most CO₂ is carried as bicarbonate ions in the plasma, with a smaller amount dissolved directly. The total amount transported within the plasma portion of 100 mL of blood is approximately 77% i.e., 3.08 mL.

The tracheae, primary, secondary and tertiary bronchi, and initial bronchioles are supported by incomplete cartilaginous rings.

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152. (3)

The Bohr effect describes the physiological phenomenon where an increase in carbon dioxide concentration (and thus an increase in hydrogen ion concentration or lower pH) leads to a decrease in haemoglobin's affinity for oxygen.

During strenuous exercise, active tissues produce more CO₂ and lactic acid, which lowers the pH. The metabolic rate also increases temperature. All these factors—high pCO₂, high H⁺ (low pH), and high temperature—destabilize the bond between oxygen and haemoglobin.

This decreased affinity for oxygen causes the oxygen dissociation curve to shift to the right.

A rightward shift means that for the same partial pressure of oxygen (e.g., at the tissue level, ~40 mm Hg), the percentage saturation of haemoglobin is lower. This is highly beneficial because it means more oxygen is released (unloaded) from the haemoglobin to meet the increased metabolic demands of the exercising tissues.

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153. (2)

The AV node receives the impulse from the SA node (in the right atrium). It introduces a ~0.1 second delay, allowing the atria to finish contracting and push blood into the ventricles. Then the impulse is sent to the Bundle of His and Purkinje fibers, causing ventricular contraction.

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154. (2)

Inspiration is initiated by the contraction of diaphragm which increases the volume of thoracic chamber in the antero-posterior axis. The contraction of external inter-costal muscles lifts up the ribs and the sternum causing an increase in the volume of the thoracic chamber in the dorso-ventral axis.

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155. (1)

Approximately 97-98.5% of the oxygen transported in the blood is chemically and reversibly bound to the haemoglobin found within red blood cells, forming oxyhaemoglobin. Only a very small fraction (1.5-3%) is transported dissolved directly in the plasma.

Oxygen is not very soluble in aqueous solutions like blood plasma. If the body had to rely solely on dissolved oxygen, the blood flow would need to be impractically high to meet even basic metabolic needs. The presence of haemoglobin in whole blood increases its oxygen-carrying capacity by about 70 times compared to plasma alone.

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156. (1)

List-I	List-II
SA Node	Located in the upper right wall of the right atrium, it generates the primary impulse for a heartbeat.
Tricuspid Valve	Prevents the backflow of blood from the right ventricle into the right atrium.
Blood Group O	Known as the "universal donor" as their RBCs lack A and B antigens.
Blood Group A	Individual has A antigens on their RBCs and can safely donate to blood groups A and AB.

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157. (3)

R–R interval = time between two consecutive ventricular depolarisations.

Given R–R interval = 1.0 sec.

$$\text{Heart rate (beats/min)} = \frac{60}{\text{R – R interval}} = \frac{60}{1.0} = 60 \text{ beats/min}$$

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158. (4)

These minerals/electrolytes are crucial for nerve function, muscle contraction, pH balance, and more. Albumin is the most abundant plasma protein and is the main contributor to the osmotic pressure that helps keep fluid within the blood vessels.

Gamma globulins are the antibodies that recognize and neutralize foreign invaders like bacteria and viruses. Platelets promote clotting, do not prevent it.

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159. (2)

Basophils are the least common type of white blood cell, making up only about 0.5-1% of the total count. The percentage 6-8% is the correct range for Monocytes.

Neutrophils – 60-65%

Monocytes – 6-8%

Lymphocytes – 20-25%

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160. (4)

The additional volume of air that can be inspired by a forcible inhalation after a normal inspiration is called the Inspiratory Reserve Volume (IRV). This volume typically averages 2500-3000 mL in a healthy adult man.

The diffusion membrane is the very thin barrier across which gas exchange occurs in the lungs. It is composed of three main layers:

- The thin squamous epithelium of the alveoli.
- The endothelium of the alveolar capillaries.
- The basement substance that lies between them.

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161. (2)

Cardiac Output (CO) = Stroke Volume (SV) × Heart Rate (HR)

Given:

CO = 5040 mL/min,

SV = 84 mL

$$\text{HR} = \frac{\text{CO}}{\text{SV}} = \frac{5040}{84} = 60 \text{ beats/min}$$

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162. (1)

Due to the continuous exchange of gases with blood, the composition of alveolar air is different from atmospheric air. It typically contains about 104 mm Hg of O₂ (approximately 14%) and 40 mm Hg of CO₂ (approximately 6%), with the remainder being mostly nitrogen.

Adult frogs are amphibians and perform gaseous exchange through multiple methods, including their lungs, skin (cutaneous respiration), and the lining of their buccal cavity (buccal respiration).

Reptilian lungs are more developed and possess a greater surface area, reflecting their more active, terrestrial lifestyle. Amphibian lungs are relatively simple, sac-like structures.

Book gills are the respiratory organs of some aquatic arthropods, such as horseshoe crabs, not reptiles. Most aquatic reptiles (like sea turtles) use lungs.

The expansion of the lungs is not caused by the rush of air. Instead, the expansion of the lungs, driven by the contraction of the diaphragm and intercostal muscles, creates a negative pressure, which causes the air to rush in.

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163. (1)

Normal, quiet expiration is a passive process because it relies on the natural relaxation of the inspiratory muscles (diaphragm and external intercostals) and the elastic recoil of the lungs and chest wall. However, forced or active expiration, such as during exercise, involves the active contraction of certain muscles to push air out more quickly and forcefully.

Passive expiration is indeed driven by the elastic recoil of the lungs after the diaphragm and external intercostal muscles relax. In contrast, active expiration is caused by the contraction of the abdominal muscles and internal intercostals, which forcefully push the diaphragm up and pull the rib cage down and in, respectively, rapidly decreasing thoracic volume.

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164. (2)

- Result with Anti-A: Agglutination occurred, which means the red blood cells have the A antigen on their surface. This eliminates O and B blood types.
- Result with Anti-B: No agglutination occurred, which means the red blood cells do not have the B antigen on their surface. This confirms the blood type is A, not AB.
- Result with Anti-Rh: No agglutination occurred, which means the red blood cells do not have the Rh antigen on their surface. This means the blood type is Rh negative.

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165. (1)

List-I	List-II
Larynx	Cartilaginous sound box
Epiglottis	Prevents entry of food in wind pipe
Carbonic anhydrase	Enzyme mainly present in RBCs
CO ₂ transport by RBC	20-25%

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166. (2)

A closed circulatory system is a system where blood is pumped by a heart through a network of closed blood vessels (arteries, veins, and capillaries) and does not come into direct contact with the body's cells.

- Annelids (segmented worms like earthworms) have a closed circulatory system.
- Chordates (all vertebrates, including fish, birds, and mammals) have a closed circulatory system.

In contrast, arthropods and most molluscs have an open circulatory system, where blood (haemolymph) is pumped into an open body cavity called a hemocoel. Sponges and coelenterates have very simple body plans and lack a circulatory system altogether.

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167. (4)

Erythroblastosis foetalis is a severe haemolytic disease of a newborn caused by Rh incompatibility between an Rh⁻ mother and an Rh⁺ foetus. The condition is prevented by administering anti-Rh antibodies (RhoGAM) to the mother, typically after the first delivery. This treatment destroys any Rh⁺ foetal red blood cells that may have entered the mother's circulation, preventing her immune system from producing its own antibodies.

Chordae tendinae are thread-like structures that attach to the flaps of the bicuspid (mitral) and tricuspid valves. However, their other ends are not attached to the auricular (atrial) wall. They are attached to the papillary muscles located on the inner surface of the ventricular wall.

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168. (3)

12-16 times per minute is the standard normal respiratory rate for a healthy adult at rest.

Insects possess a highly efficient tracheal system that delivers oxygen directly to every cell in their body, bypassing the circulatory system. This is a very direct and effective method of gas exchange compared to the respiratory systems of other invertebrates.

Atmospheric air is approximately 21% oxygen. During breathing, oxygen is absorbed by the blood, and carbon dioxide is released. As a result, exhaled air contains a lower percentage of oxygen, typically around 16%, and a higher percentage of carbon dioxide.

To protect the high-pressure flow of blood, arteries are generally located deep within the body, often alongside bones. Veins, which carry lower-pressure blood, are more numerous and often lie closer to the surface of the skin.

Birds, like mammals, have a complete 4-chambered heart (two atria and two ventricles). This separation of oxygenated and deoxygenated blood allows for double circulation, which is highly efficient for their active, high-metabolism lifestyle.

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169. (2)

In a closed circulatory system, blood is contained within a network of vessels. This confinement allows the heart to generate and maintain high blood pressure. This high pressure ensures that blood flows quickly and efficiently to all parts of the body. This rapid, targeted delivery is crucial for supporting the high metabolic demands of larger, more active organisms.

In an open circulatory system, haemolymph (the circulatory fluid) mixes directly with the interstitial fluid in the hemocoel. This mixing makes nutrient and oxygen delivery less efficient and slower.

An open circulatory system operates at a much lower pressure than a closed system. The fluid simply flows into open body cavities rather than being forced through narrow vessels.

The total volume of circulatory fluid in an open system (haemolymph) is often a large percentage of the total body volume, sometimes as much as 20-40%. While the amount of blood in a closed system is smaller relative to body volume (around 5-10%), its rapid and efficient movement makes it more effective.

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170. (2)

Expiratory reserve volume (ERV) is the additional volume of air that can be exhaled forcibly after a normal tidal expiration.

Inhale normally and then exhale maximally → this would measure tidal volume + ERV, not ERV alone.

Exhale normally and then exhale maximally → this is exactly the definition of ERV.

Inhale maximally and then exhale normally → this measures a normal tidal expiration from an inflated lung, not ERV.

Breathe normally for several cycles → this records tidal volume, not ERV.

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171. (3)

- Basophils (and their tissue-resident counterparts, mast cells) contain large granules filled with powerful chemical mediators.
- Upon activation, they release:
- Histamine: A potent vasodilator that increases blood flow and makes capillaries leaky.
- Heparin: An anticoagulant that prevents blood from clotting too quickly at the site of injury.
- Serotonin: Also contributes to the inflammatory response.
- This function is distinct from Neutrophils and Monocytes (which are phagocytic) and Eosinophils (which are primarily involved in parasitic infections and modulating allergic reactions).

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172. (1)

Basophils (a type of granulocyte in the blood) and mast cells (found in tissues) both play a key role in the inflammatory and allergic responses. They release chemicals such as histamine (a vasodilator) and heparin (an anticoagulant) at the site of injury or infection.

Phagocytosis is the process where cells engulf and digest foreign particles or pathogens. Both neutrophils and monocytes are professional phagocytes. Neutrophils are the first responders to infection, and monocytes differentiate into macrophages in tissues, which are highly effective phagocytic cells.

Specific, long-term immunity (adaptive immunity) is mediated by lymphocytes (B cells and T cells), not granulocytes. Granulocytes (basophils, eosinophils, neutrophils) are part of the innate immune system.

Erythrocytes (red blood cells) are the primary cells responsible for oxygen transport due to the presence of hemoglobin. Basophils are a type of white blood cell that deals with inflammation and allergies; they have no role in oxygen transport.

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173. (1)

List-I	List-II
Total lung capacity	5800 ml
Mask	Helps workers working in stone grinding factories
Cardiac arrest	Sudden cessation of heart beat
Coronary system	Provides nutrients and oxygen to the cardiac musculature

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174. (3)

Standard physiological values show that Residual Volume (RV) is typically around 1100-1200 mL, while Expiratory Reserve Volume (ERV) is around 1000-1100 mL. Therefore, RV is greater than ERV.

- Vital Capacity (VC) is the maximum amount of air a person can exhale after a maximum inhalation. Lung diseases like emphysema and fibrosis reduce the lungs' elasticity and/ or increase airway resistance, thereby decreasing a person's vital capacity.
- However, during quiet, regular breathing (Tidal Volume), the amount of air moved can still be within a normal range (around 500 mL) as long as the person is not undergoing any strenuous activity. The total lung capacity is affected, but the volume of normal breathing may not be.

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175. (1)

A heart attack (myocardial infarction) is an acute event where blood flow to a section of the heart muscle is suddenly blocked, most often by a blood clot. This lack of oxygen and nutrients causes that part of the heart muscle to die.

Heart failure, on the other hand, is a chronic condition where the heart is unable to pump enough blood to meet the body's needs. It's a progressive weakening of the heart's pumping function over time. A heart attack can lead to heart failure, but they are not the same condition.

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176. (4)

The solubility of CO₂ in the respiratory membrane fluid is about 20-25 times higher than that of O₂, which makes its diffusion much faster despite a smaller partial pressure gradient.

"Deoxygenated" blood returning from the tissues is typically still about 75% saturated with oxygen. It only means it has less oxygen than arterial blood.

The intercostal muscles, both external and internal, are situated in the spaces between the ribs. Their contraction and relaxation are key to changing the volume of the thoracic cavity during breathing.

Lymphocytes are of two major types – 'B' and 'T' forms.

100 ml blood = 15 g of Hb = 20 mL of O₂ i.e. 1 g of Hb = 20/15 mL of O₂ = 1.34 mL O₂

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177. (2)

The P-wave represents the depolarization (electrical signal for contraction) of the **atria** (upper chambers).

The QRS complex represents the depolarization of the **ventricles** (lower chambers), which is the electrical event that leads to their strong contraction.

The T-wave represents the repolarization (electrical reset or relaxation) of the **ventricles**.

Heart rate is determined by measuring the time between two consecutive R-waves (the peaks of the QRS complexes).

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178. (3)

The spleen is a major lymphatic organ with a crucial role in blood filtration. It acts as a "graveyard" for red blood cells, removing old, damaged, and non-functional erythrocytes from circulation. The spleen's narrow blood vessels and specialized macrophages filter and destroy these senescent cells.

Without the spleen, this primary filtering mechanism is lost. As a result, the old and damaged red blood cells that would normally be removed by the spleen remain in the bloodstream, leading to an increased number of circulating senescent erythrocytes over time.

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179. (1)

The Bundle of His is located in the interventricular septum, not the auriculoventricular (atrial-ventricular) septum. It's a key part of the heart's electrical conduction system that transmits impulses from the AV node to the ventricles.

Papillary muscles are attached to the atrioventricular (AV) valves via the chordae tendinae. They contract during ventricular systole to prevent the valve flaps from flipping back into the atria.

Under resting, normal physiological conditions, every 100 mL of oxygenated blood delivers approximately 5 mL of oxygen to the tissues.

Each haemoglobin can carry maximum four molecules at a time.

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180. (3)

When oxygen (O_2) binds to the iron atom in haemoglobin, it is a process of oxygenation, where the molecule is simply combined with oxygen. It is not an oxidation process, which would involve the loss of electrons and a change in the iron's valency. The binding of oxygen to haemoglobin is a readily reversible process. This is why haemoglobin can pick up oxygen in the lungs and release it in the tissues. Crucially, the iron atom in the haem group remains in its ferrous (Fe^{2+}) state. If it were oxidized to the ferric (Fe^{3+}) state, it would form methaemoglobin, which cannot bind and release oxygen, making the process irreversible.

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